

openQCM Next Impedance Analysis: measuring shifts in frequency and bandwidth based on calculation of conductance G

A Additional and Supporting Information

A1. The impedance circuit modification of the AD8302 INPB circuit branch

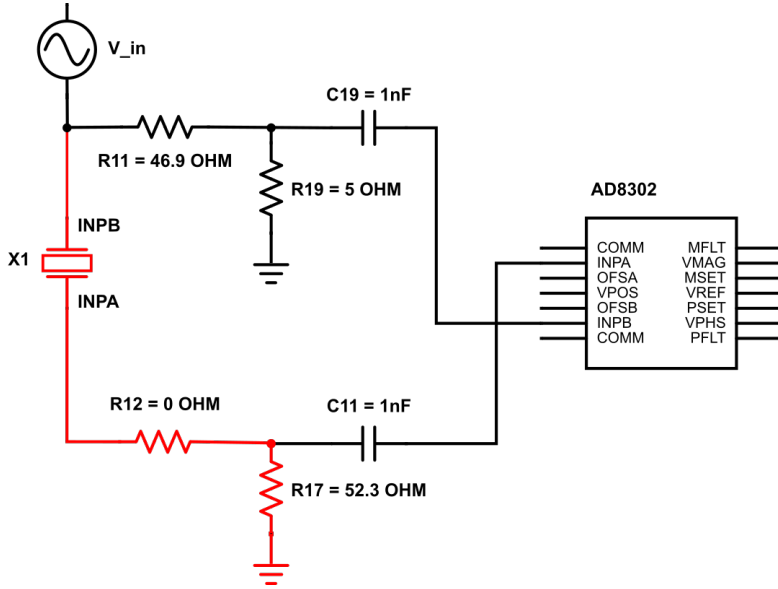


Figure A 1.1. openQCM Next voltage divider circuit, highlighted in red, and modification to the INPB circuit branch

In the openQCM Next circuit the INPB circuit branch was modified according to the observation reported below:

Resistors R11 and R19 in the INPB branch has been changed. Possible values for R11 are between 39 and 47 Ohm. Possible values for R19 are 5 and 10 Ohm. Basically, we need the sum of R11 and R19 to be close to 50 Ohm and R11/R19 to be between 5 and 10. One of the problems with AD8302 was, the INPA/INPB is too small when the crystal has fluid on top. So small that, it is pushed to the nonlinear regime. We are trying to lower INPB 5-10 times to carry the INPA/INPB to the linear region. (by Zehra Parlak, Postdoctoral Associate, Mechanical Engineering and Materials Science & Biochemistry, Duke University)

By changing resistors R11 and R19, the voltage V_{INPB} fed to the AD8302 is lower than the actual voltage stimulating the quartz crystal sensor, called V_{IN} . It is a voltage divider circuit whose transfer function is given by:

$$H = \frac{V_{INPB}}{V_{IN}} = \frac{R_{19}}{R_{11} + R_{19}} \Rightarrow$$

$$V_{INPB} = \frac{R_{19}}{R_{11} + R_{19}} V_{IN} = \frac{5\Omega}{46.9\Omega + 5\Omega} V_{IN} \Rightarrow \quad (A1.1)$$

$$V_{INPB} \cong \frac{V_{IN}}{10}$$

In conclusion, changing the resistors R11 and R19 in this way lowered the actuating voltage V_{IN} of a factor 10, approximately. It is necessary to consider relationship 1.1 when calculating the ratio $\frac{V_{IN}}{V_{INA}}$ from the measured gain loss $V_{MAG}(\text{misured})$ output of AD8302 component. The actual ration ratio $\frac{V_{IN}}{V_{INA}}$ is calculated as:

$$V_{MAG}(misured) = R_F I_{SLP} \log\left(\frac{V_{INA}}{V_{INB}}\right) + V_{CP}, using (1.1) \Rightarrow$$

$$V_{MAG}(misured) = R_F I_{SLP} \left(\log\left(10 \times \frac{V_{INA}}{V_{IN}}\right) \right) + V_{CP}$$

doing some math the ratio $\frac{V_{IN}}{V_{INA}}$ is given by:

$$\frac{V_{IN}}{V_{INA}} = 10^{\left(\frac{V_{CP} + R_F I_{SLP} - V_{MAG}}{R_F I_{SLP}}\right)} \quad (A1.2)$$

Note that the quantity $\frac{V_{IN}}{V_{INA}}$ in equation 1.2 is the actual transfer function of the voltage divider circuit formed by the QCM sensor and the resistor R17 connected to ground (Figure 1 highlighted in red) and it is used to calculate the conductance G

A2. Processing of AD8302 raw data for the calculation of Conductance $G(V_{MAG}, V_{PHS})$

The AD8302 output voltage V_{MAG} and V_{PHS} raw data were analyzed for the calculation of conductance G using the algorithm described below:

1. Correction of the V_{MAG} signal by removing the “baseline signal” measured on the entire frequency range from 1MHz to 50 MHz
2. Savitzky-Golay filtering of the V_{MAG} and V_{PHS} raw data
3. Interpolation and smoothing of V_{MAG} and V_{PHS} filtered signals using spline-fitting and [UnivariateSpline](#) python class
4. Calculation of quartz impedance modulus $|Z_q|$, according to equation:

$$|Z_q|(V_{MAG}) = R_{17} \left(10^{\left(\frac{V_{CP} + R_F I_{SLP} - V_{MAG}}{R_F I_{SLP}}\right)} - 1 \right) \quad (A2.1)$$

5. Calculation of quartz impedance phase ϕ according to equation:

$$\phi(V_{PHS}) = \frac{V_{CP} - V_{PHS}}{R_F I_{\phi}} + 90^\circ \quad (A2.2)$$

6. Calculation of the conductance G , the real part of admittance, as:

$$G(V_{MAG}, V_{PHS}) = \frac{\cos \phi(V_{PHS})}{|Z_q|(V_{MAG})} \quad (A2.3)$$

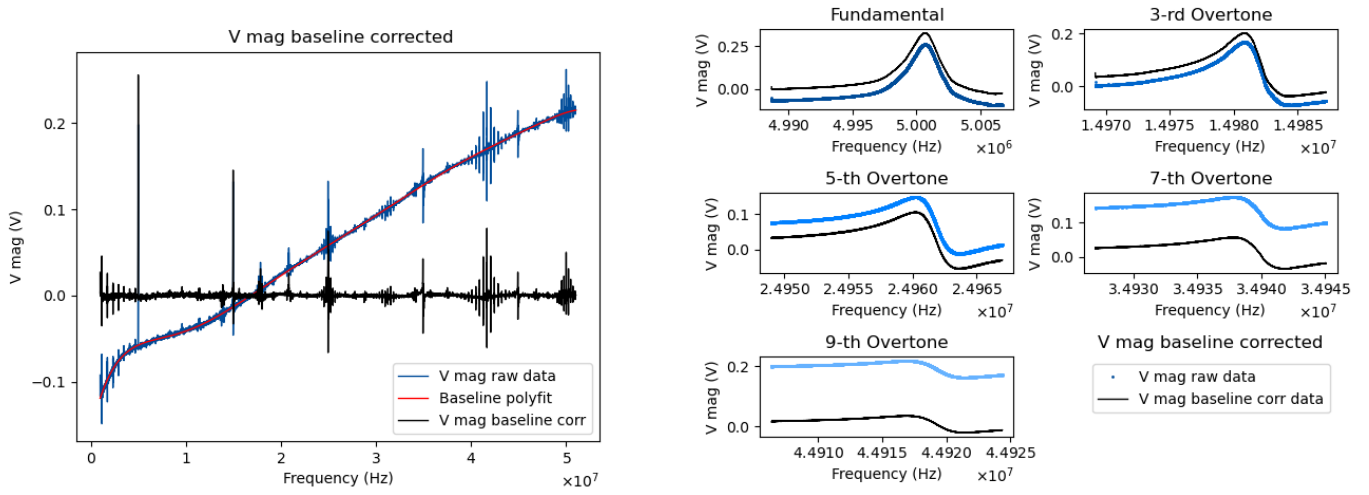


Figure A2.1 Left: V_{MAG} raw data baseline estimation and correction over the entire sweep frequency range from 1 MHz to 50 MHz; Right: Correction of the V_{MAG} signal by removing the “baseline signal”.

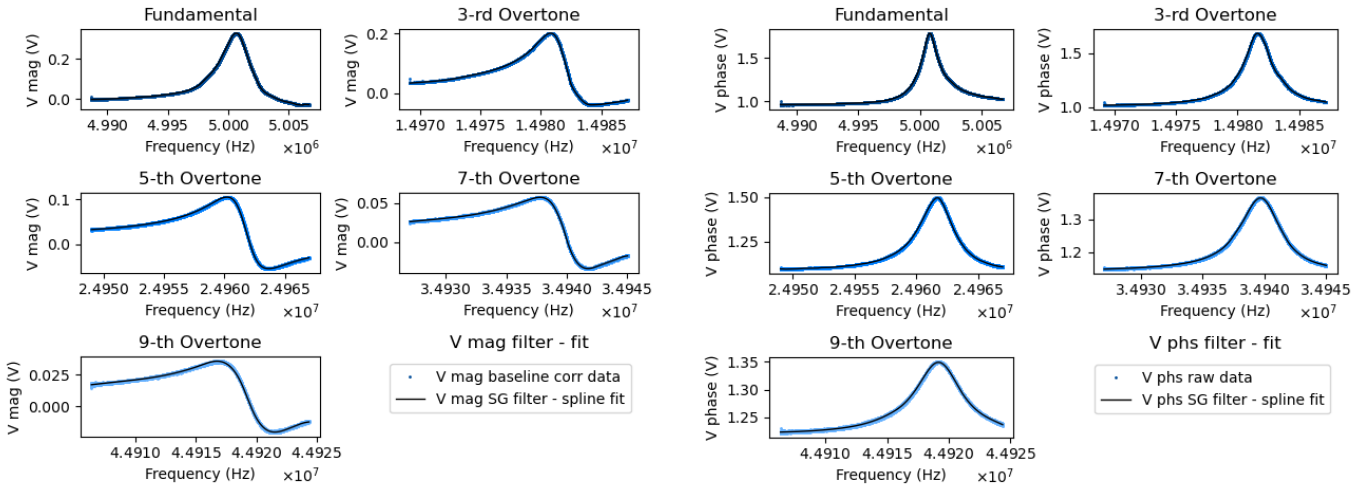


Figure A2.2 Left: Savitzky-Golay filtering and spline interpolation - smoothing of V_{MAG} ; Right: Savitzky-Golay filtering and spline interpolation - smoothing of V_{PHS}

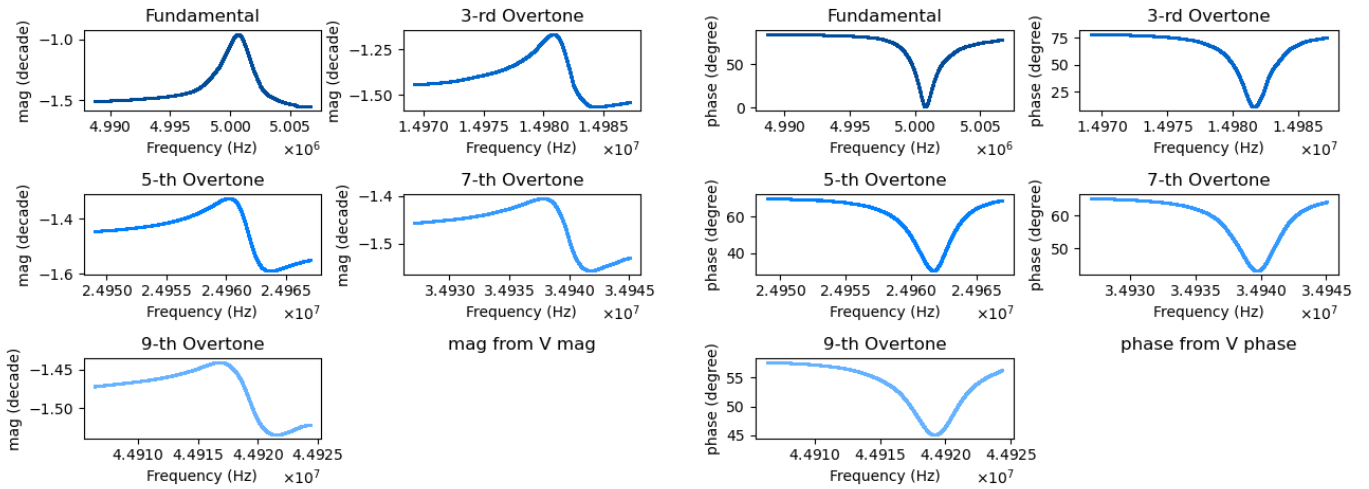


Figure 3.3 Left: magnitude $\log \frac{V_{IN}}{V_{INA}}$ in decade derived from V_{MAG} and Right: phase ϕ in degree derived from V_{PHS} according to the transfer equation of the AD8302

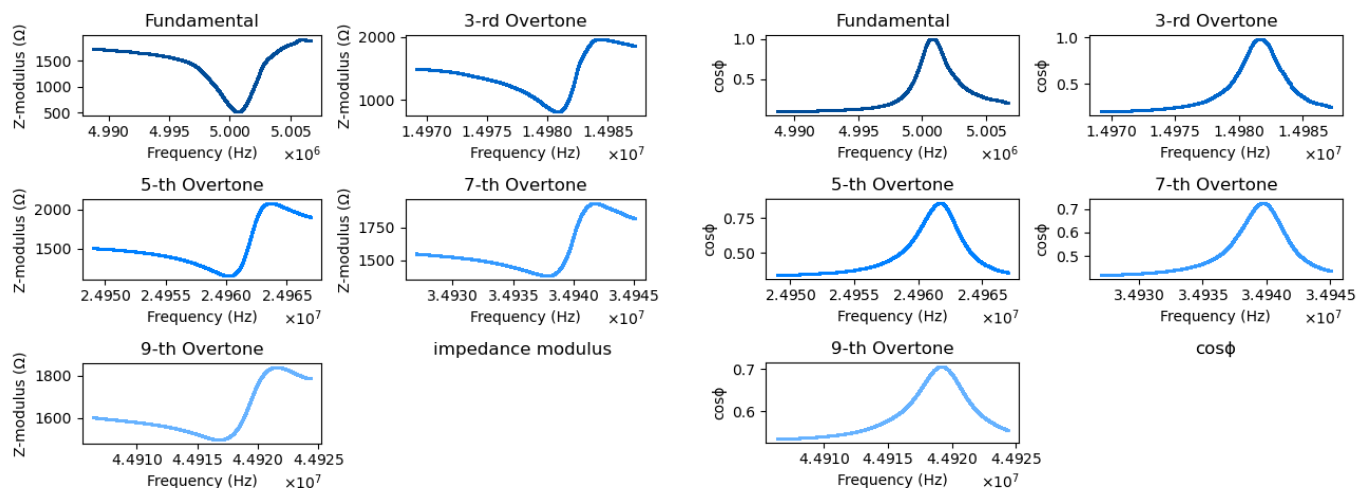


Figure A2.3 Left: quartz impedance modulus $|Z_q|$ and Right: $\cos \phi$ of the quartz impedance phase

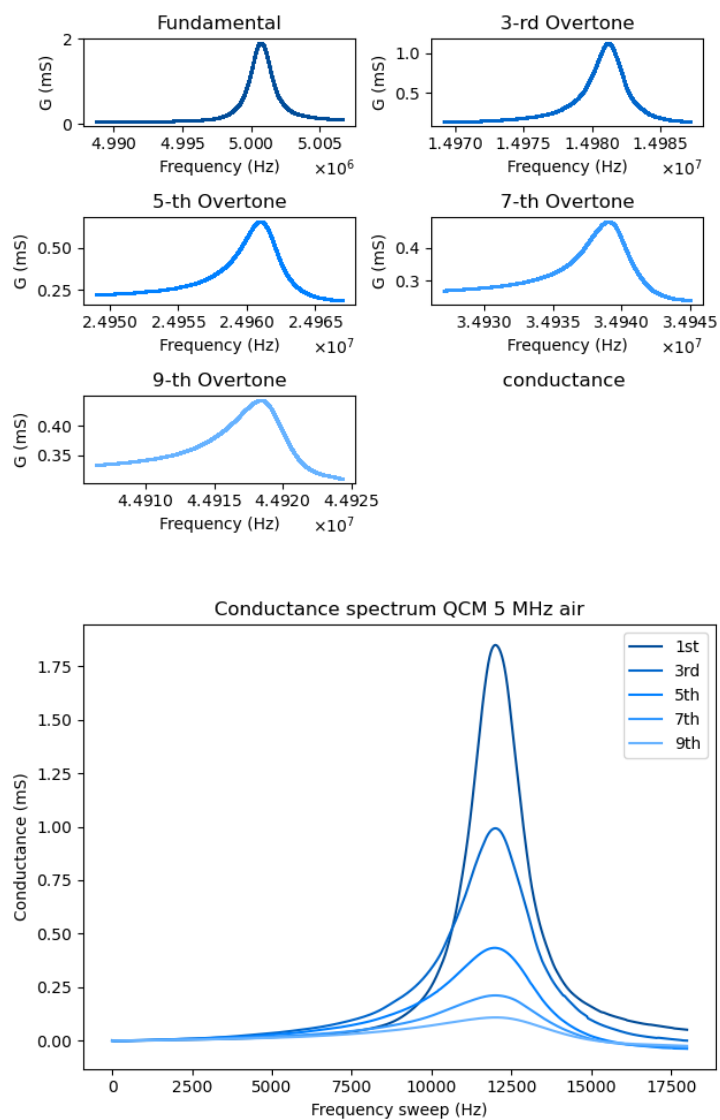
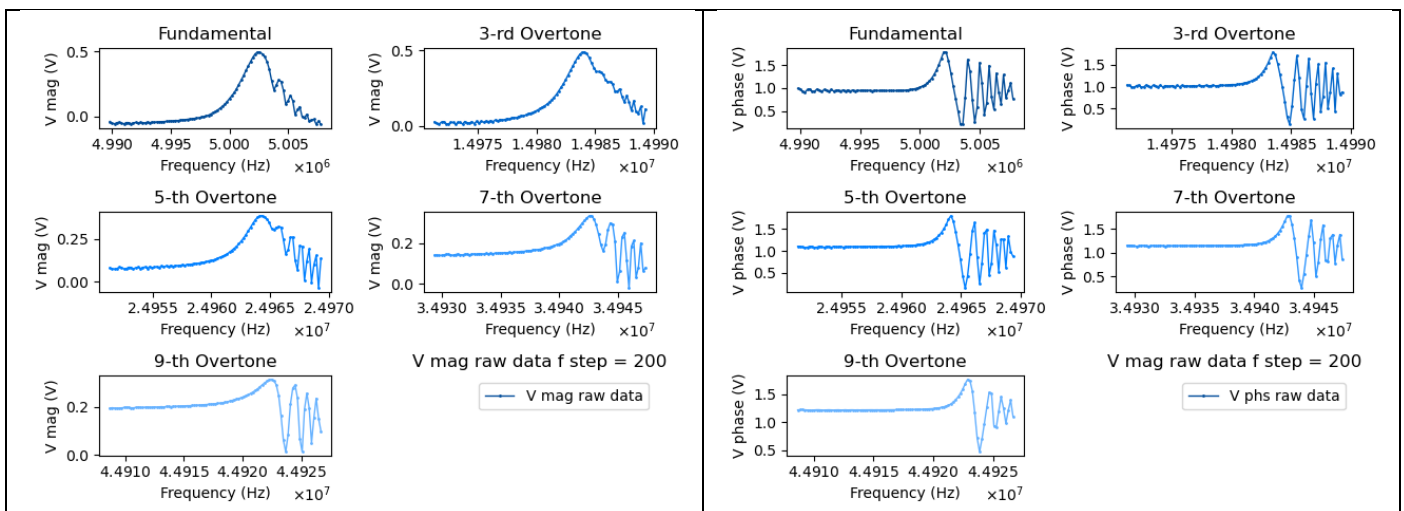
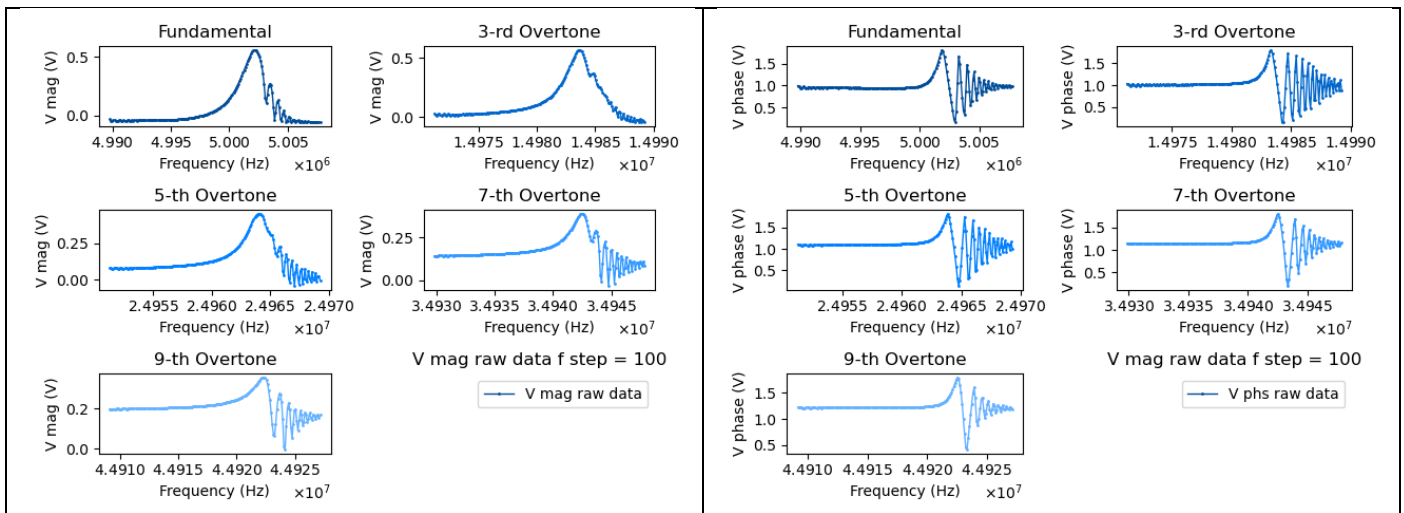
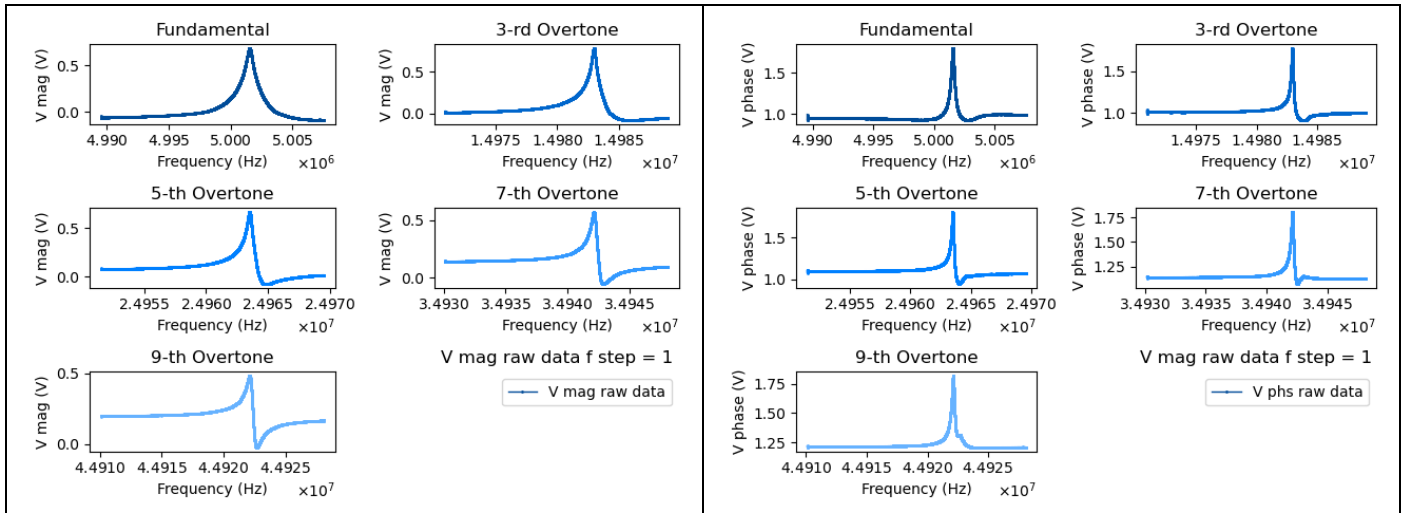


Figure A2.4 Impedance spectrum of 5 MHz Au-coated 14 mm diameter quartz sensor in contact with water.

A3. Sweep Rate Dependency AD8302 Signal acquisition and ADC measurement



A4. Susceptance and Admittance Circle

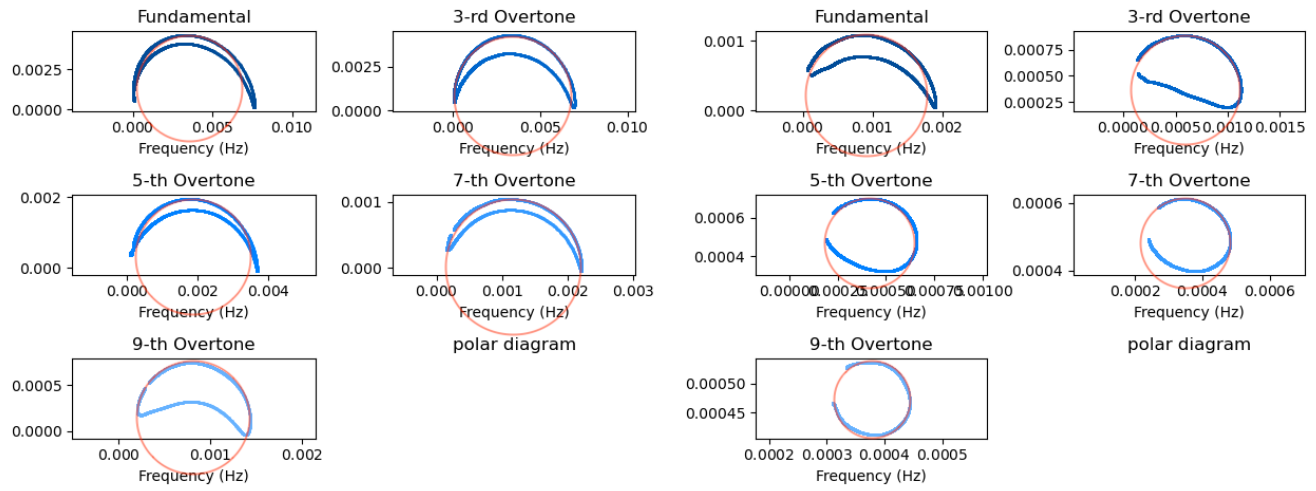


Figure A4.1

Polar diagram Susceptance B vs conductance G for quartz sensor in contact air (left) and water (right). Ideally in polar diagram will result in a circular form